Full INS/GNSS Reference Filter Mathematical Formulation

sensor_fusion documentation

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1 Scope

This document describes the Full INS/GNSS filter implemented in `sensor_fusion/src/full/` and `sensor_fusion/src/full/generated.rs`. It is used for replay, diagnostics, and comparison with the runtime Reduced. It carries ECEF navigation states, IMU bias and scale correction states, and the physical vehicle-to-body mount used by inertial propagation.

2 Frames and Conventions

Symbol	**Meaning**
e	ECEF frame. Full nominal position and velocity are propagated in this frame.
n	Local NED frame. GNSS standard deviations may be supplied in this frame and are rotated into ECEF.
b	Raw IMU body/sensor frame. Full IMU deltas are supplied in this frame.
v	Vehicle frame, forward-right-down. NHC is expressed in this frame.

Rotations follow the project convention

$$x_a = C_{ab} x_b, \qquad R(q_{ab}) = C_{ab}, \qquad R(q_1 \otimes q_2) = R(q_1)R(q_2). \tag{1}$$

The full attitude quaternion q_{ev} is the ECEF-frame attitude with respect to the vehicle frame. Its DCM maps vehicle-frame coordinates into ECEF coordinates:

$$C_{ev} = R(q_{ev}), \qquad x_e = C_{ev} x_v. \tag{2}$$

The mount quaternion stored in the Rust fields `qcs0..qcs3` is the physical vehicle-to-body mount q_{bv}:

$$C_{bv} = R(q_{bv}), \qquad x_b = C_{bv} x_v, \qquad C_{vb} = C_{bv}^T, \qquad x_v = C_{vb} x_b. \tag{3}$$

The local NED-to-ECEF relation is represented by

$$x_n = C_{ne} x_e, \qquad x_e = C_{ne}^T x_n. \tag{4}$$

3 Nominal State

The nominal state contains 26 scalar fields:

$$x = \begin{bmatrix} q_{ev} & v_e^T & p_e^T & b_g^T & b_a^T & s_g^T & s_a^T & q_{bv} \end{bmatrix}^T. \tag{5}$$

The public Rust fields are:

Fields	Meaning
`q0..q3`	q_{ev}, vehicle frame to ECEF attitude
`vn,ve,vd`	ECEF velocity v_e, meters per second
`pn,pe,pd`	ECEF position p_e, meters
`bgx..bgz`	gyro additive correction b_g, radians per second
`bax..baz`	accelerometer additive correction b_a, meters per second squared
`sgx..sgz`	gyro scale correction s_g
`sax..saz`	accelerometer scale correction s_a
`qcs0..qcs3`	physical vehicle-to-body mount quaternion q_{bv}

The code applies scale and additive correction directly to IMU rates and specific force:

$$\omega_b^c = s_g \odot \omega_{\mathrm{raw},b} + b_g, \tag{6}$$
$$f_b^c = s_a \odot f_{\mathrm{raw},b} + b_a. \tag{7}$$

Consequently, these are correction states. A physical sensor bias appears with the opposite sign of the additive correction required to remove it.

4 Error State

The 24-dimensional error state order in the Rust update and generated matrices is

$$\delta x = \begin{bmatrix} \delta p_e & \delta v_e & \delta\theta_v & \delta b_a & \delta b_g & \delta s_a & \delta s_g & \delta\psi_{bv} \end{bmatrix}^T. \tag{8}$$

Index ranges are:

Indices	Error component
0..2	ECEF position error δp_e
3..5	ECEF velocity error δv_e
6..8	vehicle-attitude small angle $\delta\theta_v$
9..11	accelerometer-bias correction error δb_a
12..14	gyro-bias correction error δb_g
15..17	accelerometer-scale error δs_a
18..20	gyro-scale error δs_g
21..23	vehicle-to-body mount error $\delta\psi_{bv}$

5 Nominal Propagation

The full filter uses a two-sample IMU increment. For each half-step $i \in \{1, 2\}$,

$$\omega_{b,i}^c = s_g \odot \left(\frac{\Delta\theta_{b,i}}{\Delta t}\right) + b_g, \tag{9}$$
$$f_{b,i}^c = s_a \odot \left(\frac{\Delta v_{b,i}}{\Delta t}\right) + b_a, \tag{10}$$
$$\omega_{v,i} = C_{vb}\omega_{b,i}^c, \tag{11}$$
$$f_{v,i} = C_{vb} f_{b,i}^c, \qquad C_{vb} = C_{bv}^T. \tag{12}$$

The attitude derivative includes Earth rotation:

$$\dot{q}_{ev} = \frac{1}{2} q_{ev} \otimes \begin{bmatrix} 0 & \omega_v^T \end{bmatrix}^T + \frac{1}{2}\begin{bmatrix} \omega_{ie}q_3 & \omega_{ie}q_2 & -\omega_{ie}q_1 & -\omega_{ie}q_0 \end{bmatrix}^T. \tag{13}$$

Specific force is rotated into ECEF:

$$f_e = C_{ev} f_v. \tag{14}$$

Velocity dynamics use the J2 ECEF gravity model and Coriolis terms:

$$\dot{v}_e = f_e + g_e(p_e) - 2\,\Omega_{ie}^e \times v_e. \tag{15}$$

The implementation writes this as component terms using $\omega_{ie} = 7.292115 \times 10^{-5}$ rad/s. Position obeys

$$\dot{p}_e = v_e. \tag{16}$$

The nominal propagation uses a predictor-corrector/trapezoidal structure: evaluate derivatives at the current state, form a temporary state after Δt, evaluate derivatives again, then average the two derivatives for the final position, velocity, and quaternion update. The nominal bias, scale, and mount states are constant during propagation.

6 Error Propagation

The generated full model provides

$$\delta x_{k+1} = F_k \delta x_k + G_k w_k, \qquad P^-_{k+1} = F_k P^+_k F^T_k + G_k Q_k G^T_k. \tag{17}$$

The 21-noise vector groups are accelerometer white noise, gyro white noise, accelerometer-bias random walk, gyro-bias random walk, accelerometer-scale random walk, gyro-scale random walk, and mount random walk:

$$w = \begin{bmatrix} n_a & n_g & n_{ba} & n_{bg} & n_{sa} & n_{sg} & n_m \end{bmatrix}^T. \tag{18}$$

The runtime diagonal entries are

$$Q_k = \operatorname{diag}\left(\sigma_a^2 \Delta t I_3,\, \sigma_g^2 \Delta t I_3,\, \sigma_{ba}^2 \Delta t I_3,\, \sigma_{bg}^2 \Delta t I_3,\, \sigma_{sa}^2 \Delta t I_3,\, \sigma_{sg}^2 \Delta t I_3,\, \sigma_m^2 \Delta t I_3\right). \tag{19}$$

7 GNSS Reference Measurements

7.1 Position

GNSS position is supplied in ECEF:

$$z_p = p_e + \eta_p. \tag{20}$$

When a horizontal accuracy h_{acc} is supplied, the implementation forms a local NED diagonal covariance

$$R_n = \operatorname{diag}\left(h_{\text{acc}}^2,\, h_{\text{acc}}^2,\, (2.5 h_{\text{acc}})^2\right), \tag{21}$$

rotates it to ECEF, then applies a Cholesky whitening transform T_p. The actual rows fused are

$$r_p = T_p(z_p - \hat{p}_e), \qquad H_p = T_p \begin{bmatrix} I_3 & 0 & \cdots & 0 \end{bmatrix}. \tag{22}$$

7.2 Velocity

GNSS velocity is supplied in ECEF, with either a scalar speed accuracy or per-axis NED velocity standard deviations. In the per-axis case, the NED velocity covariance is rotated to ECEF and Cholesky-whitened:

$$r_v = T_v(z_v - \hat{v}_e), \qquad H_v = T_v \begin{bmatrix} 0 & I_3 & 0 & \cdots & 0 \end{bmatrix}. \tag{23}$$

If only scalar speed accuracy is available, the three ECEF velocity rows use that scalar variance directly.

8 Non-Holonomic Constraint

The full NHC predicts vehicle-frame velocity from ECEF velocity:

$$v_v = C_{ev}^T v_e. \tag{24}$$

The lateral and vertical constraints are

$$h_y = e_y^T v_v \approx 0, \qquad h_z = e_z^T v_v \approx 0. \tag{25}$$

The generated rows include derivatives with respect to velocity and vehicle attitude. They have no direct mount block because mount affects NHC through propagation-induced covariance, not through the instantaneous NHC measurement:

$$\frac{\partial(e_i^T v_v)}{\partial\delta\psi_{bv}} = 0, \qquad i \in \{y, z\}. \tag{26}$$

NHC rows are gated by speed and a quasi-static IMU check:

$$\|v_e\| \geq 0.5 \text{ m/s}, \qquad \|\omega_b\| < 0.2 \text{ rad/s}, \qquad |\|f_b\| - 9.81| < 1.0 \text{ m/s}^2. \tag{27}$$

A scalar chi-square gate prevents large residuals from being fused.

9 Batch Update and Injection

The full filter batches up to eight rows: three position rows, three velocity rows, and two NHC rows. It uses a sequential covariance update in f64 shadow storage:

$$S_i = H_i P H_i^T + R_i, \tag{28}$$
$$K_i = P H_i^T S_i^{-1}, \tag{29}$$
$$\delta x \leftarrow \delta x + K_i(r_i - H_i \delta x), \tag{30}$$
$$P \leftarrow P - K_i S_i K_i^T. \tag{31}$$

After the batch, the accumulated error is injected:

$$p_e^+ = p_e^- + \delta p_e, \qquad v_e^+ = v_e^- + \delta v_e, \tag{32}$$
$$q_{ev}^+ = \delta q(\delta\theta_v) \otimes q_{ev}^-, \tag{33}$$
$$b_a^+ = b_a^- + \delta b_a, \qquad b_g^+ = b_g^- + \delta b_g, \tag{34}$$
$$s_a^+ = s_a^- + \delta s_a, \qquad s_g^+ = s_g^- + \delta s_g, \tag{35}$$
$$q_{bv}^+ = \delta q(\delta\psi_{bv}) \otimes q_{bv}^-. \tag{36}$$

Both quaternions are normalized. After injection, the attitude and mount covariance blocks are reset with the generated first-order quaternion reset Jacobian

$$G_r(\delta\theta) \simeq I - \frac{1}{2}[\delta\theta]_\times. \tag{37}$$

The reset is applied to the q_{ev} attitude block and to the q_{bv} mount block, then the covariance is symmetrized.

10 ECEF/NED Initialization

For visualizer replay, a vehicle yaw in the local NED frame is converted to the ECEF-frame attitude with respect to the vehicle frame q_{ev}, using latitude and longitude. The physical mount q_{bv} is then initialized from Align, reference data, or a manual caller-supplied mount. Raw body-frame IMU samples are supplied directly to the Full filter.

11 Observability and Known Limitations

- GNSS position and velocity directly observe only ECEF position and velocity. Attitude, biases, scales, and mount are updated through covariance coupling and inertial dynamics.
- Lateral and vertical NHC rows do not directly observe mount. Mount is updated through propagation-induced covariance with attitude and velocity.
- Accelerometer vertical bias and accelerometer vertical scale are strongly coupled in ground-vehicle motion dominated by gravity. The visualizer commonly keeps accelerometer scale fixed for diagnostics.
- The additive bias states are correction states. Diagnostic plots should not be interpreted as physical sensor bias without applying the sign convention.
- Mount, attitude, gyro bias, and gyro scale can become correlated during turns. Compare vehicle attitude and physical mount separately when diagnosing mount convergence.